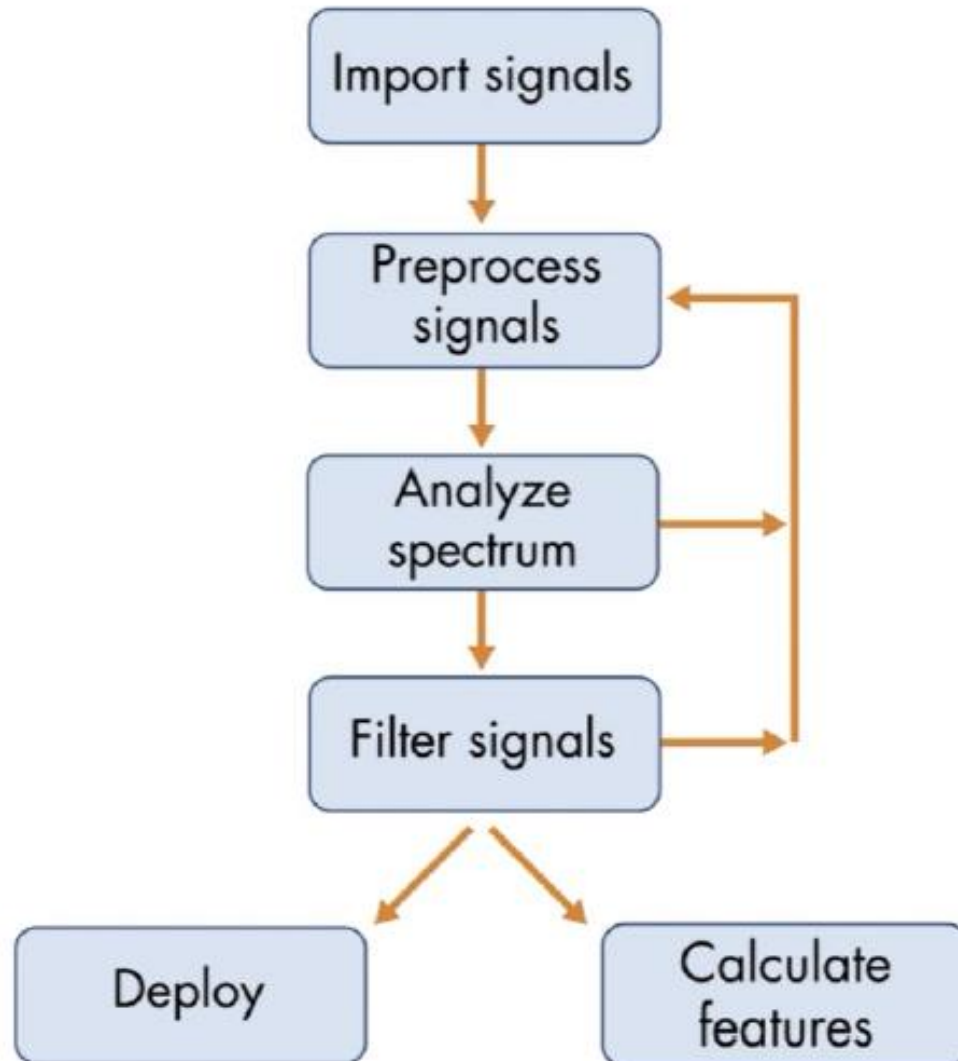


Signal Processing Using MATLAB

24th Sep, 2023

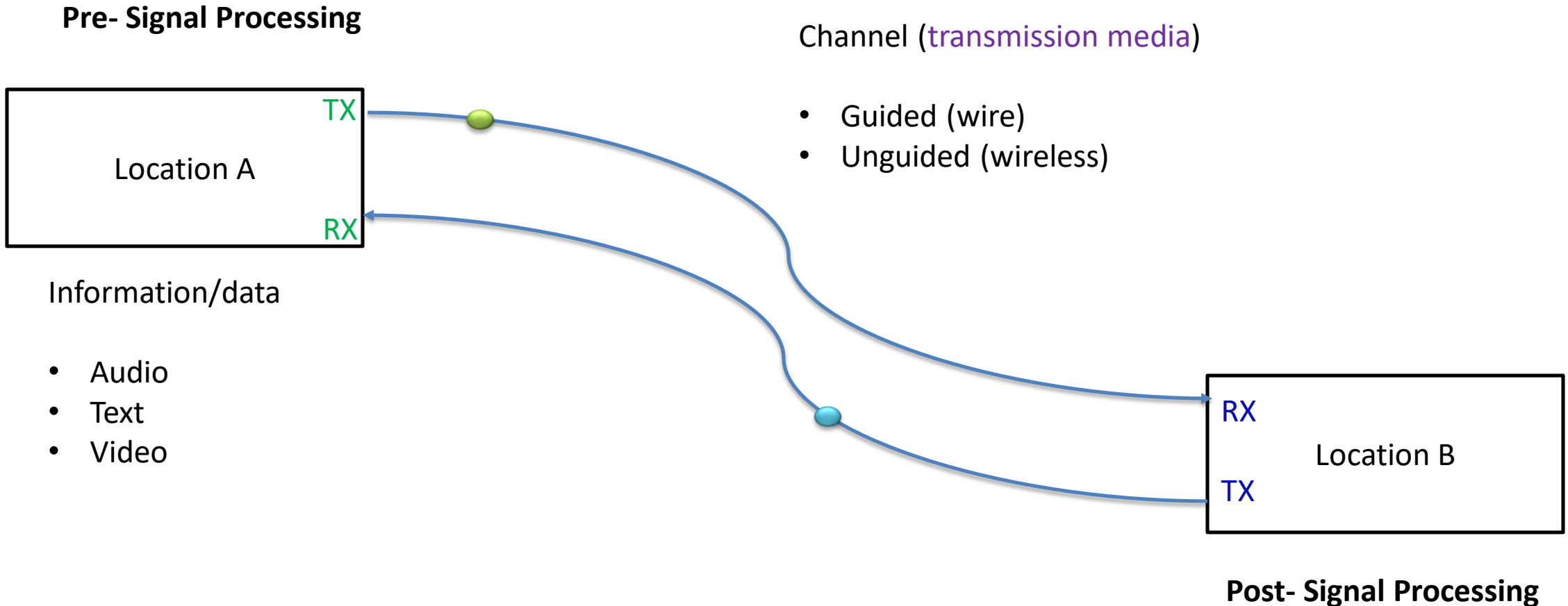
Dr. Saif Ali Jasim Al Grait

Overview: Signal Processing Workflow



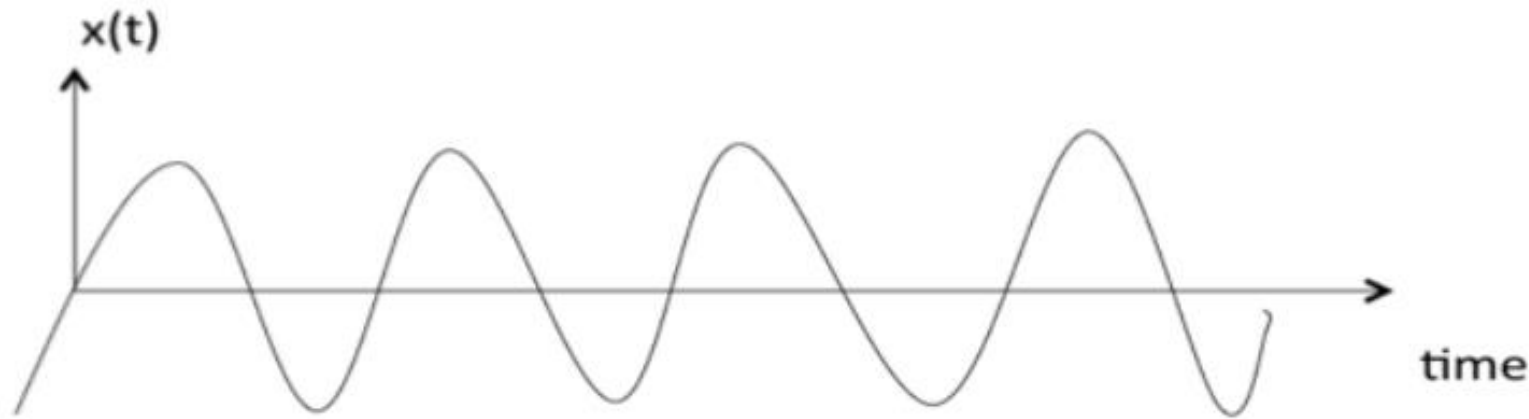
1. Import the signals into MATLAB
2. Preprocess the signals in the time domain to prepare them for analysis
3. Look at the power spectrum in the frequency domain
4. Filter the signals to keep only relevant frequency content
5. Revisit earlier steps in the process to refine your algorithm
6. Extract information from the signals or deploy your algorithm to hardware

Communication: Exchange of information; Transfers information from one place to another



A signal is a function representing a physical quantity or variable

- “A Signal contains information about the behavior or nature of the phenomenon
- Mathematically, a signal is represented as a function of an independent variable ‘t’
- Usually ‘t’ represents time. Thus, a signal is denoted by **$x(t)$** ”



Signals types can be classified into the following categories:

- Continuous Time and Discrete Time Signals.
- Deterministic and Non-deterministic Signals.
- Even and Odd Signals.
- Periodic and Aperiodic Signals.
- Energy and Power Signals.
- Real and Imaginary Signals.

Unit Step Function

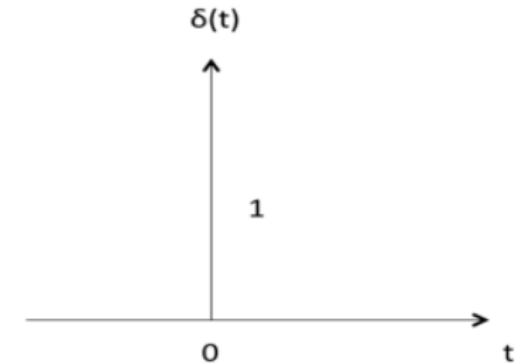
Unit step function is denoted by $u(t)$. It is defined as $u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$



- It is used as best test signal.
- Area under unit step function is unity.

Unit Impulse Function

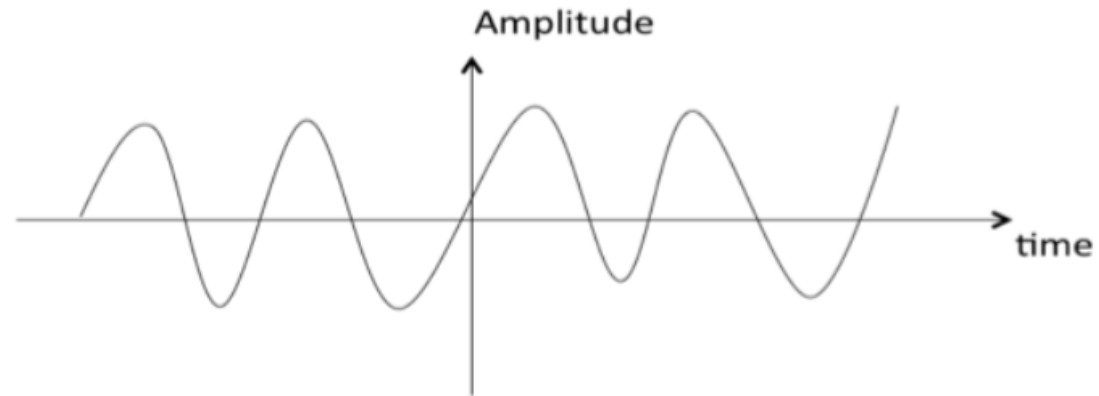
Impulse function is denoted by $\delta(t)$. and it is defined as $\delta(t) = \begin{cases} 1 & t = 0 \\ 0 & t \neq 0 \end{cases}$



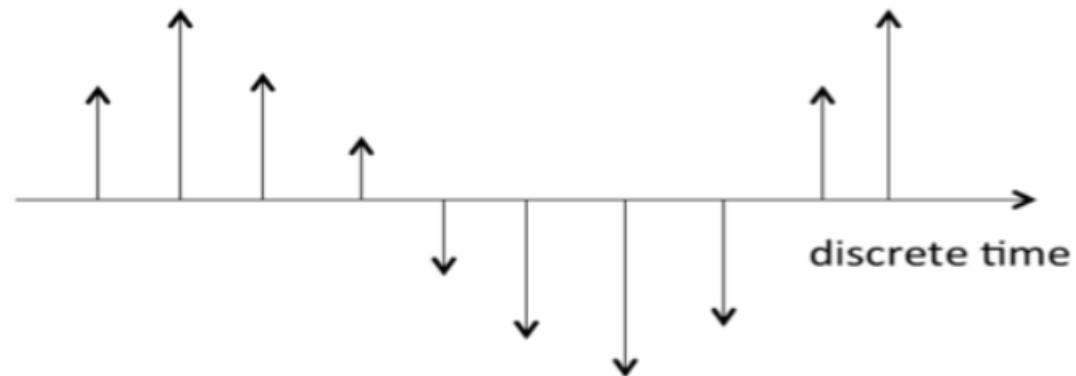
Continuous vs discrete signals in time

Continuous Time and Discrete Time Signals

A signal is said to be continuous when it is defined for all instants of time.

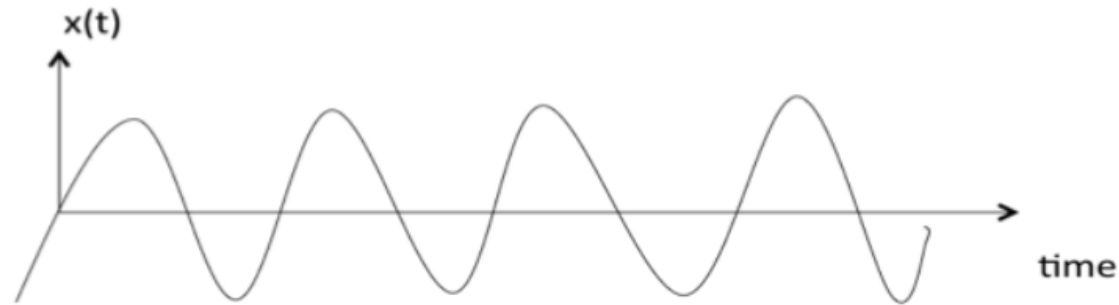


A signal is said to be discrete when it is defined at only discrete instants of time/

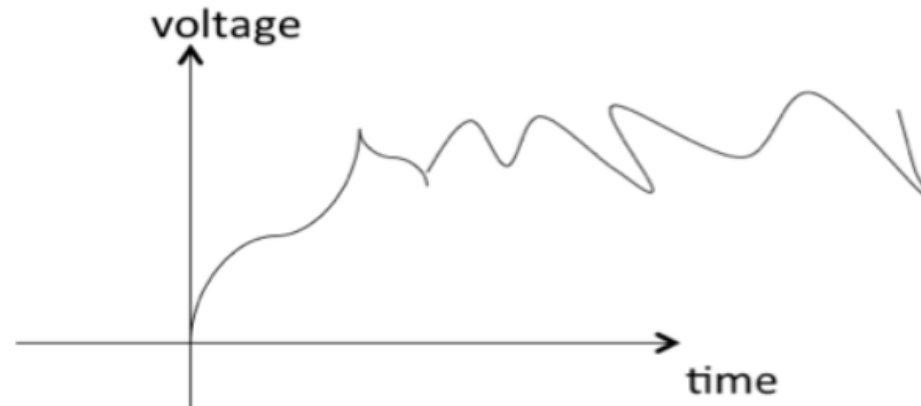


Deterministic and Non-deterministic Signals

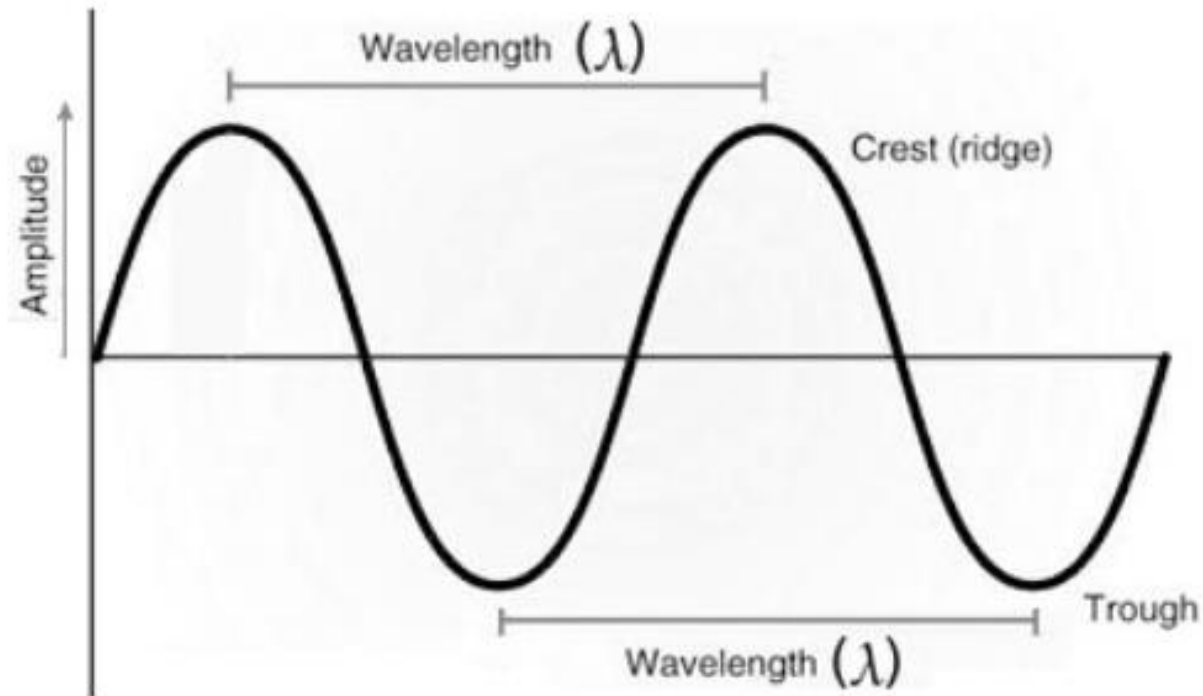
A signal is said to be deterministic if there is no uncertainty with respect to its value at any instant of time. Or, signals which can be defined exactly by a mathematical formula are known as deterministic signals.



A signal is said to be non-deterministic if there is uncertainty with respect to its value at some instant of time. Non-deterministic signals are random in nature hence they are called random signals. Random signals cannot be described by a mathematical equation. They are modelled in probabilistic terms.



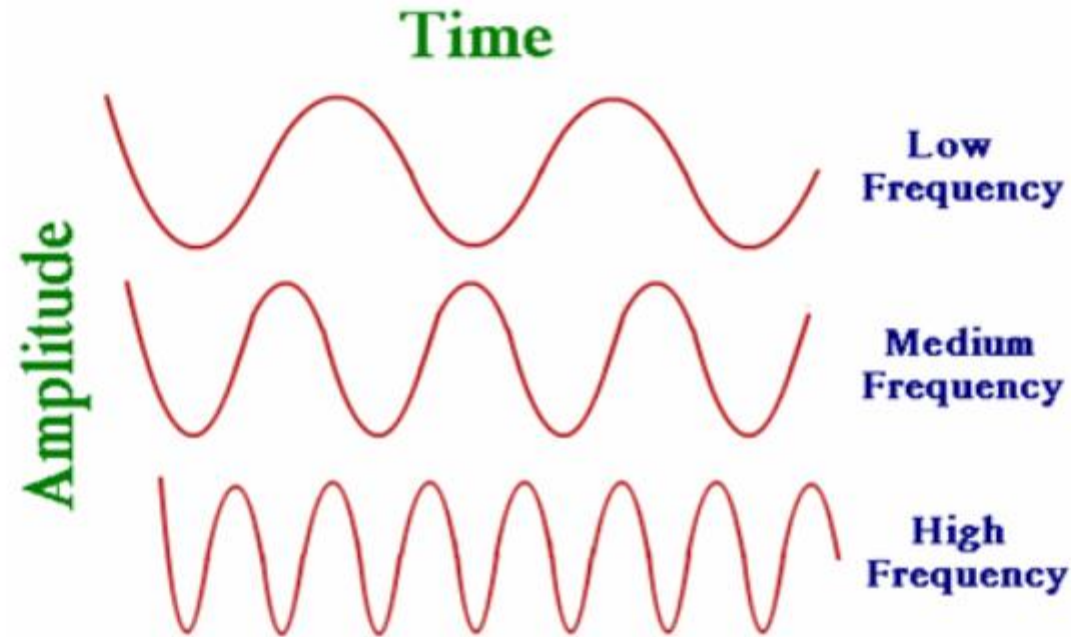
Signal property can be characterized based on amplitude, frequency and phase



Frequency

What is Frequency?

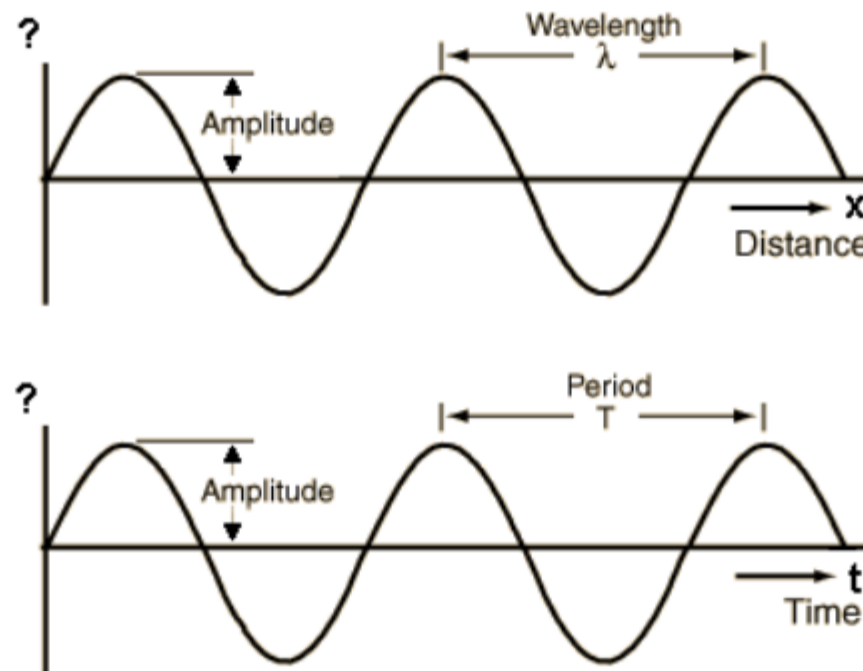
The frequency of a waveform is the number of times a complete waveform is repeated in a fixed time period. The unit of frequency is Hz which corresponds to the number of waveforms that repeat in 1 second. So if a wave repeats 5 times in one second its frequency would be 5 Hz.



Wavelength

What is Wavelength?

The wavelength of a wave is the distance between two adjacent peaks or troughs of adjacent cycles. It is represented by lambda (λ) and has units of distance like meters, cm, mm etc. The wavelength applies to both traveling waves and standing waves. The Wavelength is inversely proportional to frequency.



Relationship between frequency and wavelength

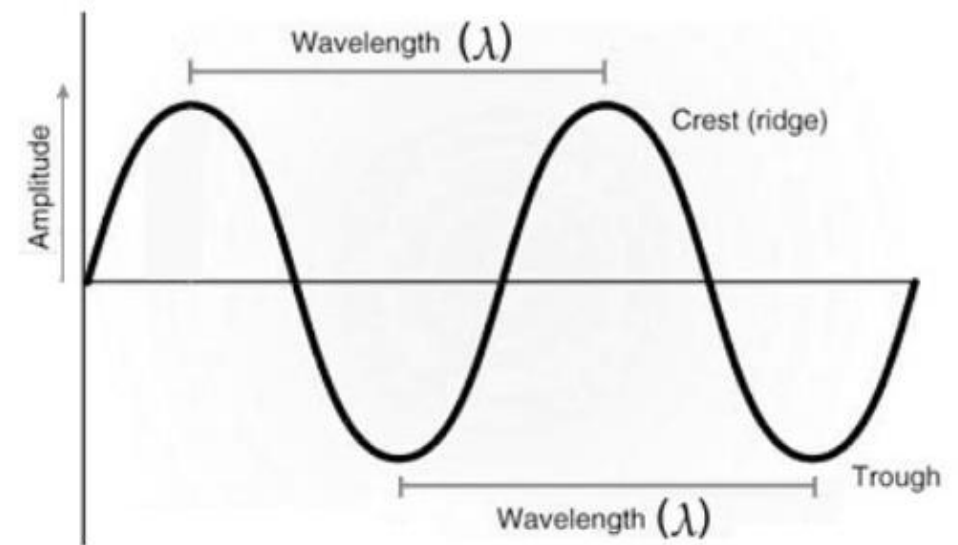
Formula: $\lambda = C/f$

Where,

λ (Lambda) = Wavelength in meters

c = Speed of Light (299,792,458 m/s)

f = Frequency



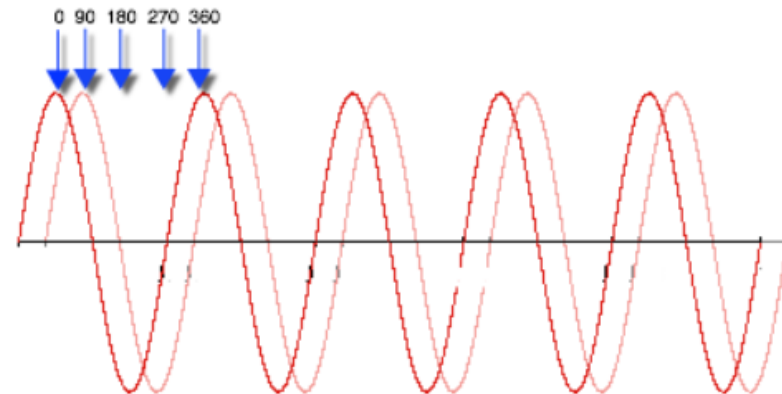
Phase

Phase is the same frequency, same cycle, same wavelength, but are 2 or more wave forms not exactly aligned together.

"Phase is not a property of just one RF signal but instead involves the relationship between two or more signals that share the same frequency. The phase involves the relationship between the position of the amplitude crests and troughs of two waveforms."

Phase can be measured in distance, time, or degrees. If the peaks of two signals with the same frequency are in exact alignment at the same time, they are said to be in phase. Conversely, if the peaks of two signals with the same frequency are not in exact alignment at the same time, they are said to be out of phase." - The CWNA definition of Phase v106

Below is an example of 2 wave forms 90 degree out of phase.



Data Vs Information

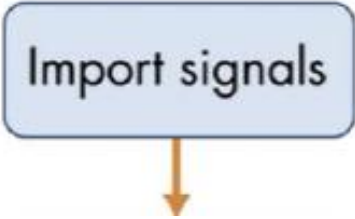
What is the difference between data & information?

Data	Information
• Data refers to raw facts that have no specific meaning.	• Information refers to processed data that has a purpose and meaning.
• The word 'data' is derived from the Latin word 'datum', which means 'something that is given'.	• The word 'information' is derived from the Latin word 'informatiō', which means 'formation or conception'.
• The data is independent of the information.	• Information is dependent on data.
• Data or raw data is not enough to make a decision.	• The information is sufficient to help make a decision in the respective context.

Ref: [\[1\]](#)

Import signal: using **readmatrix** function in MATLAB

Import signals



1. Import the signals into MATLAB

Syntax

`A = readmatrix(filename)`

`A = readmatrix(filename,opts)`

`A = readmatrix(___,Name,Value)`

Read matrix from file

Example: Read file.csv

```
A = readmatrix("A.csv")  
plot(A)
```

https://www.mathworks.com/help/matlab/ref/readmatrix.html#mw_020aafa6-861b-42e8-bc54-7c71b32a6066

Import signal: using **readtimetable** function in MATLAB

“Instead of storing your signal in a numeric variable and keeping track of the time information separately, you can use a timetable. A timetable is a table where each sample has an associated time”

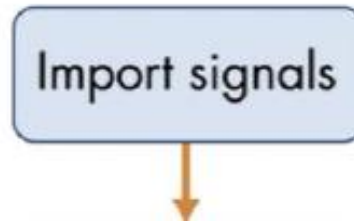
Syntax

```
TT = readtimetable(filename)
```

```
TT = readtimetable(filename,opts)
```

```
TT = readtimetable(___,Name,Value))
```

Create timetable from file

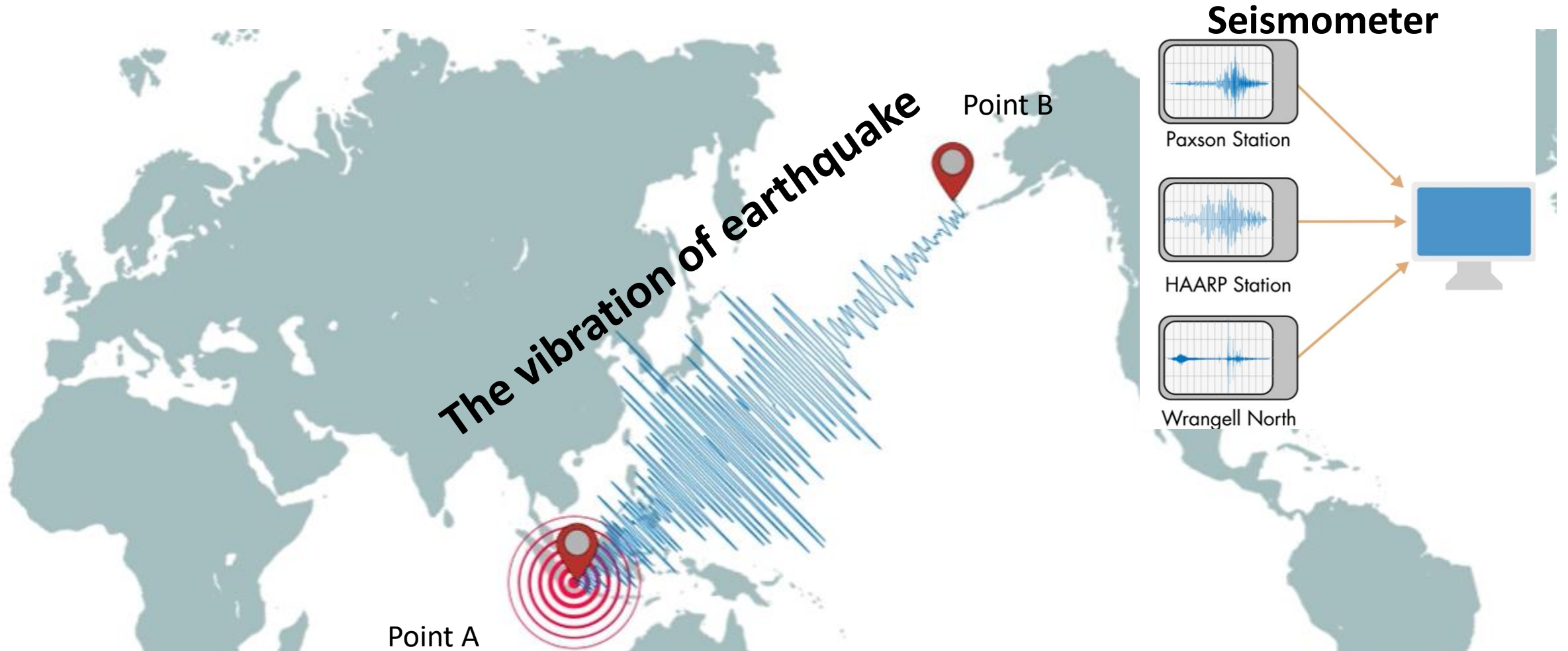


Example: Read file.csv

```
TT = readtimetable('outages.csv');
```

<https://www.mathworks.com/help/matlab/ref/readtimetable.html>

Task 1: Perform signal processing for earthquake



Seismic Waves

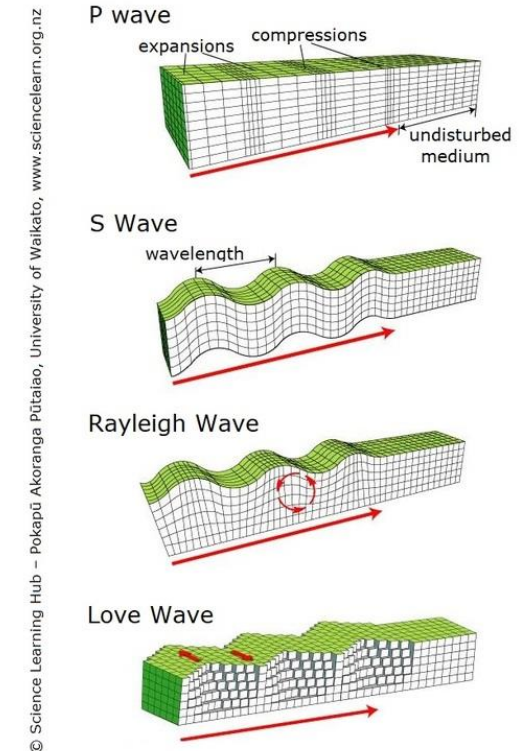
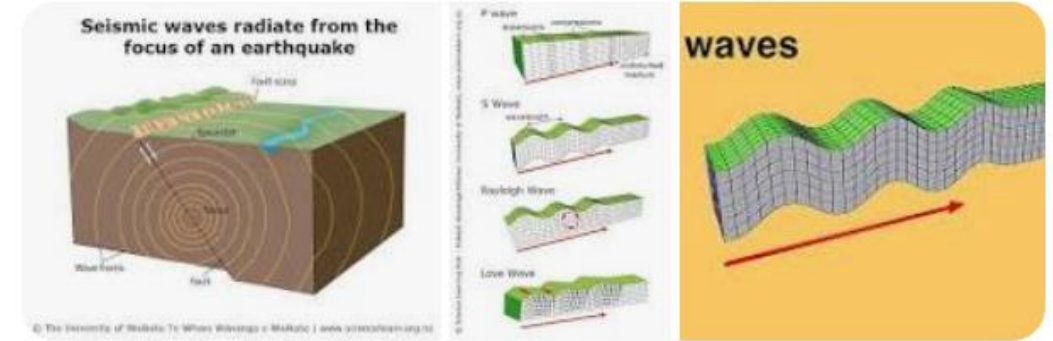
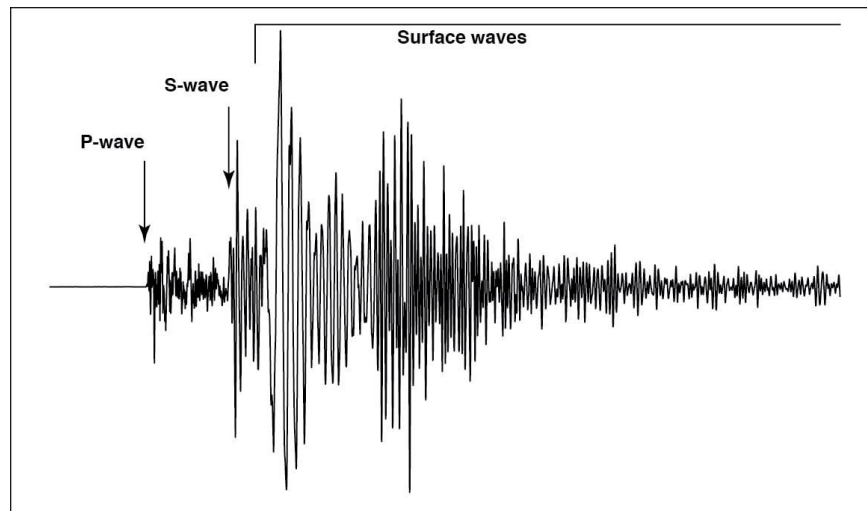
Seismic waves are caused by the sudden movement of materials within the Earth, such as slip along a fault during an earthquake. Volcanic eruptions, explosions, landslides, avalanches, and even rushing rivers can also cause seismic waves.



Michigan Technological University

<https://www.mtu.edu/geology/learn/seismology-study>

Seismology | UPSeis | Michigan Tech



Step 1) Import Signal into MATLAB

The sample rate of the `harp` signal is 50 Hz. Since the sample rate is the reciprocal of the time step, this means the time in between each sample is 0.02

seconds: $\frac{1}{50\text{Hz}}$.

Instead of storing your signal in a numeric variable and keeping track of the time information separately, you can use a timetable. A timetable is a table where each sample has an associated time.

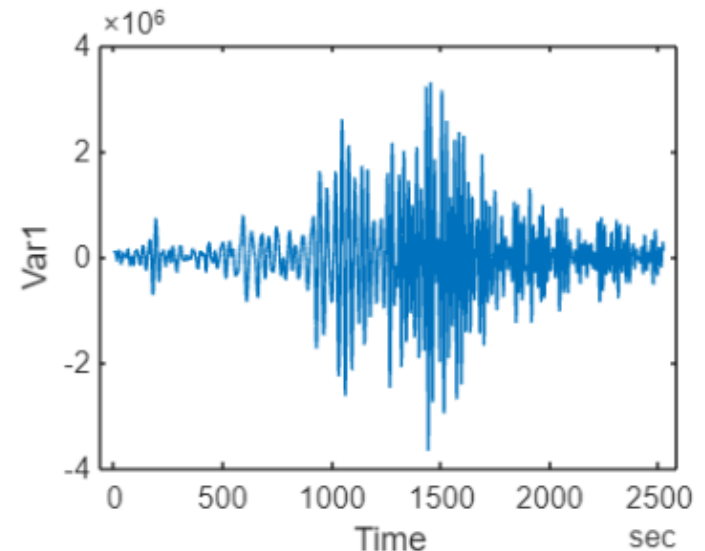
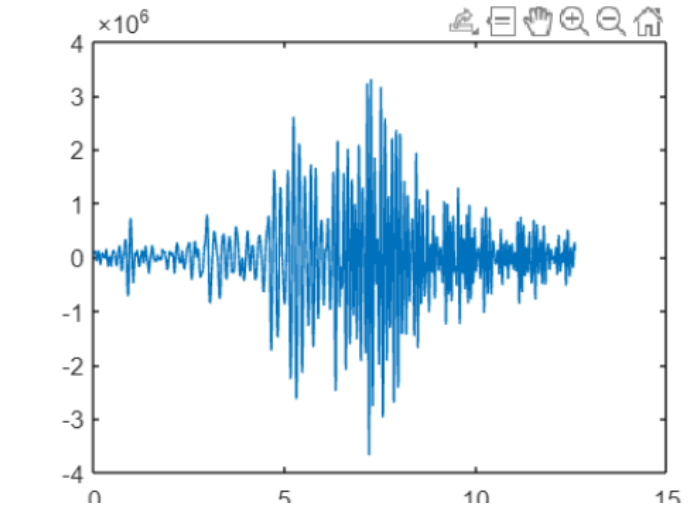
You can import data into a timetable with the `readtimetable` function. Use the `"SampleRate"` option to calculate the time for each sample.

```
tbl = readtimetable("harp.csv", ...  
    "SampleRate",50)  
plot(tbl,"Var1")
```

Workspace			
Name	Value	Size	Class
harp	126000×1 double	126000×1	double
tbl	126000×1 timetable	126000×1	timetable

harp = 126000×1		
	1	
1		65252
2		65556
3		65837
4		66141
5		66453
6		66758
7		67067
8		67371

tbl = 126000×1 timetable		
	Time	Var1
1	0 sec	65252
2	0.02 sec	65556
3	0.04 sec	65837
4	0.06 sec	66141
5	0.08 sec	66453
6	0.1 sec	66758
7	0.12 sec	67067
8	0.14 sec	67371
9	0.16 sec	67694

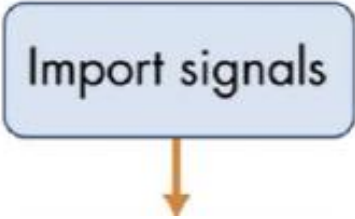


Step 1) Import Signal into MATLAB

Workspace			
Name	Value	Size	Class
 harp	126000×1 double	126000×1	double
 tbl	126000×1 timetable	126000×1	timetable

Import signal: using **readmatrix** function in MATLAB

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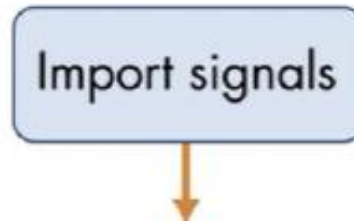
Syntax

```
TT = readtimetable(filename)
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TT = readtimetable(filename,opts)
```

```
TT = readtimetable(___,Name,Value))
```

Create timetable from file



Example: Read file.csv

```
TT = readtimetable('outages.csv');
```

<https://www.mathworks.com/help/matlab/ref/readtimetable.html>

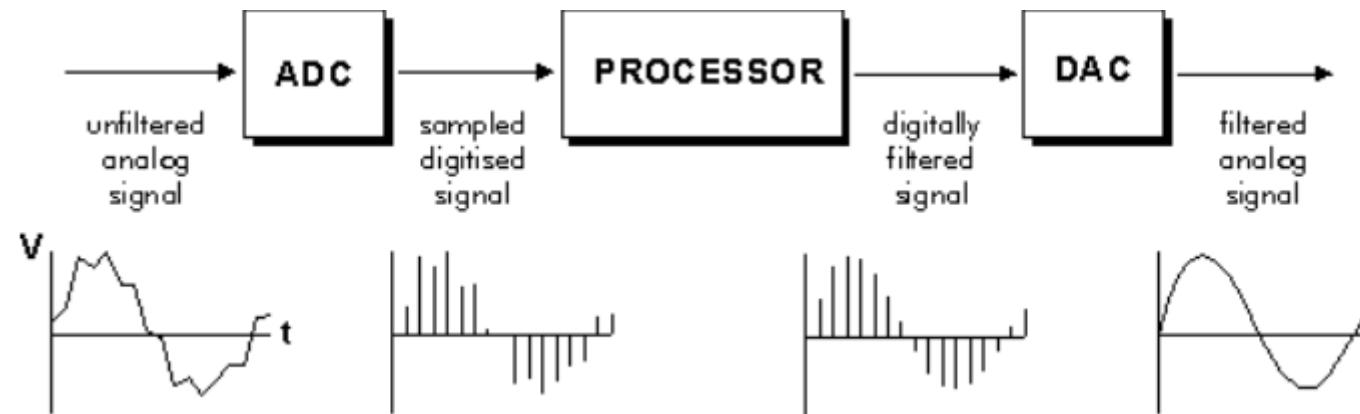
Signal processing Concept

- “Signal processing consists of mapping or transforming information bearing signals into another form of signals at the output, aiming at some application benefits”



Signal processing Concept

- Mapping could involve analog to digital conversion to perform processing in digital domain



Ref: [projectspune](http://projectspune.com)

Signal processing Concept

- Transformation
- Filtering
- Detection
- Estimation
- Recognition and classification
- Coding
- Analyzing

Signal processing Concept

- Transformation
- Filtering
- Detection
- Estimation
- Recognition and classification
- Coding
- Analyzing

Spectrum Analyzer in MATLAB

is an app for inspecting, comparing and signal processing

Signal Analyzer

Visualize and compare multiple signals and spectra

Description

The Signal Analyzer app is an interactive tool for visualizing, preprocessing, measuring, analyzing, and comparing signals in the time domain, in the frequency domain, and in the time-frequency domain. Using the app, you can:

- Easily access all the signals in the MATLAB® workspace
- Smooth, filter, resample, detrend, denoise, duplicate, extract, rename, and edit signals without leaving the app
- Add and apply custom preprocessing functions
- Visualize and compare multiple waveform, spectrum, persistence, spectrogram, and scalogram representations of signals simultaneously
- Measure data and signal statistics

The Signal Analyzer app provides a way to work with many signals of varying durations at the same time and in the same view.

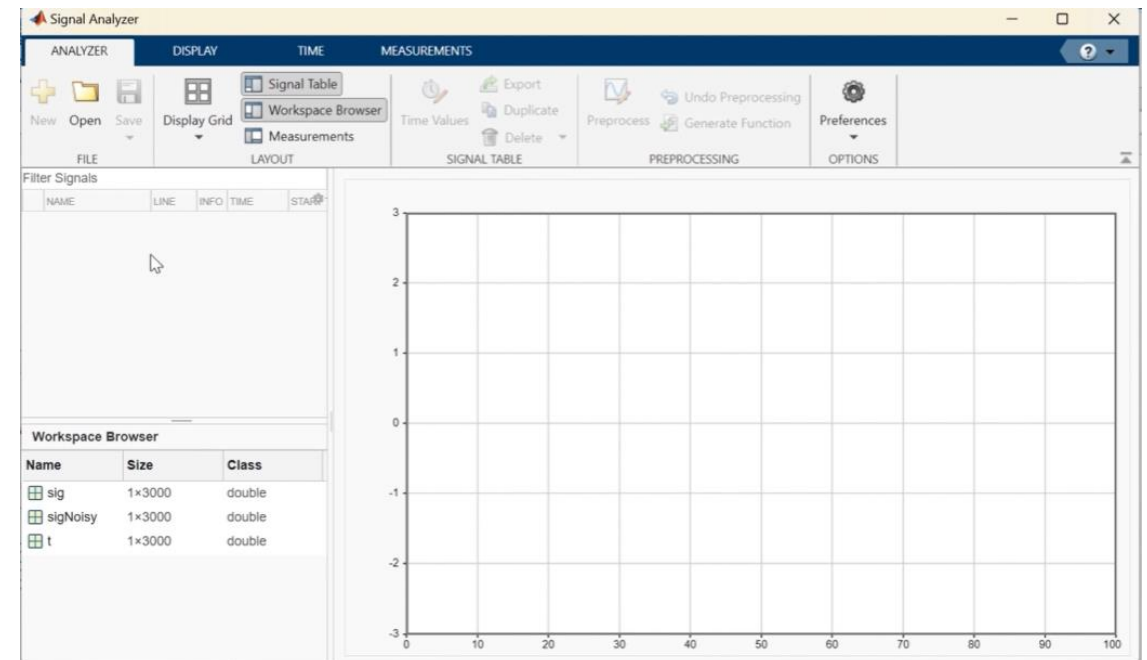
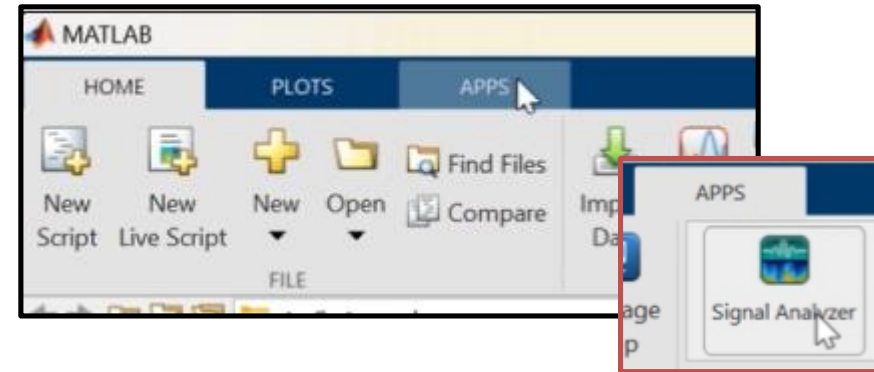
For more information, see [Using Signal Analyzer App](#).

<https://www.mathworks.com/help/signal/ref/signalanalyzer-app.html>

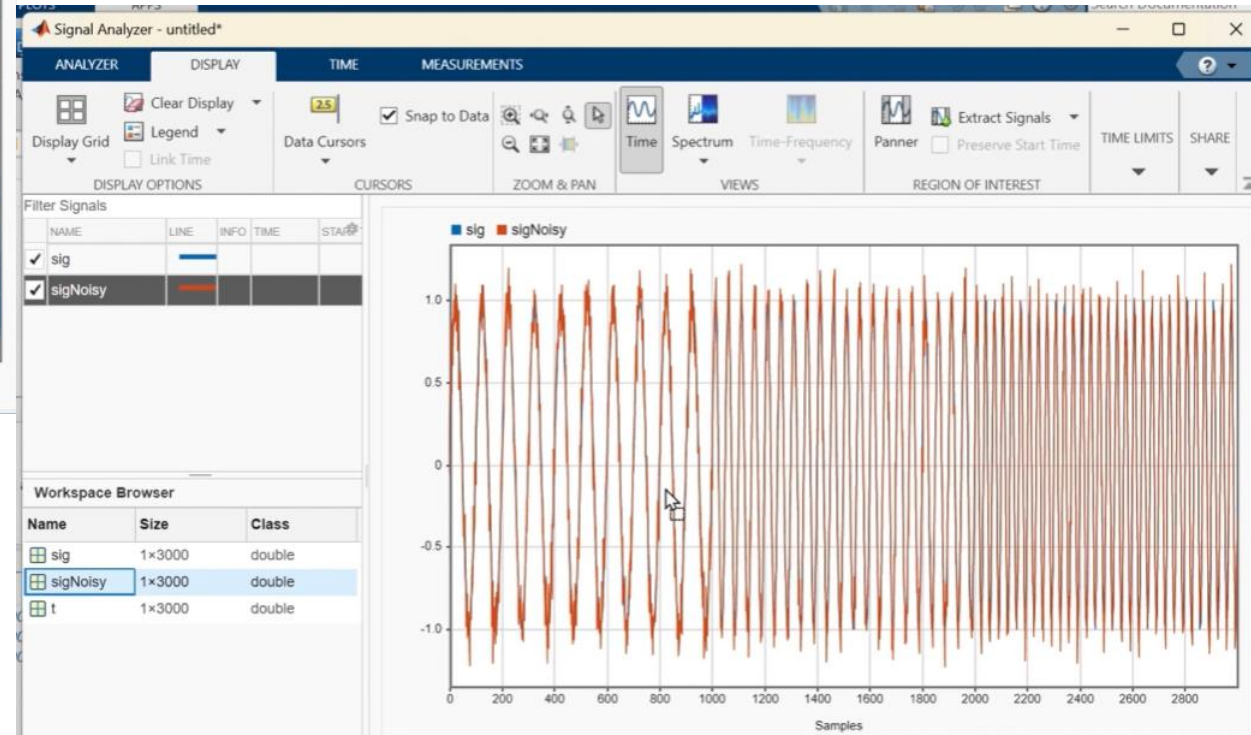
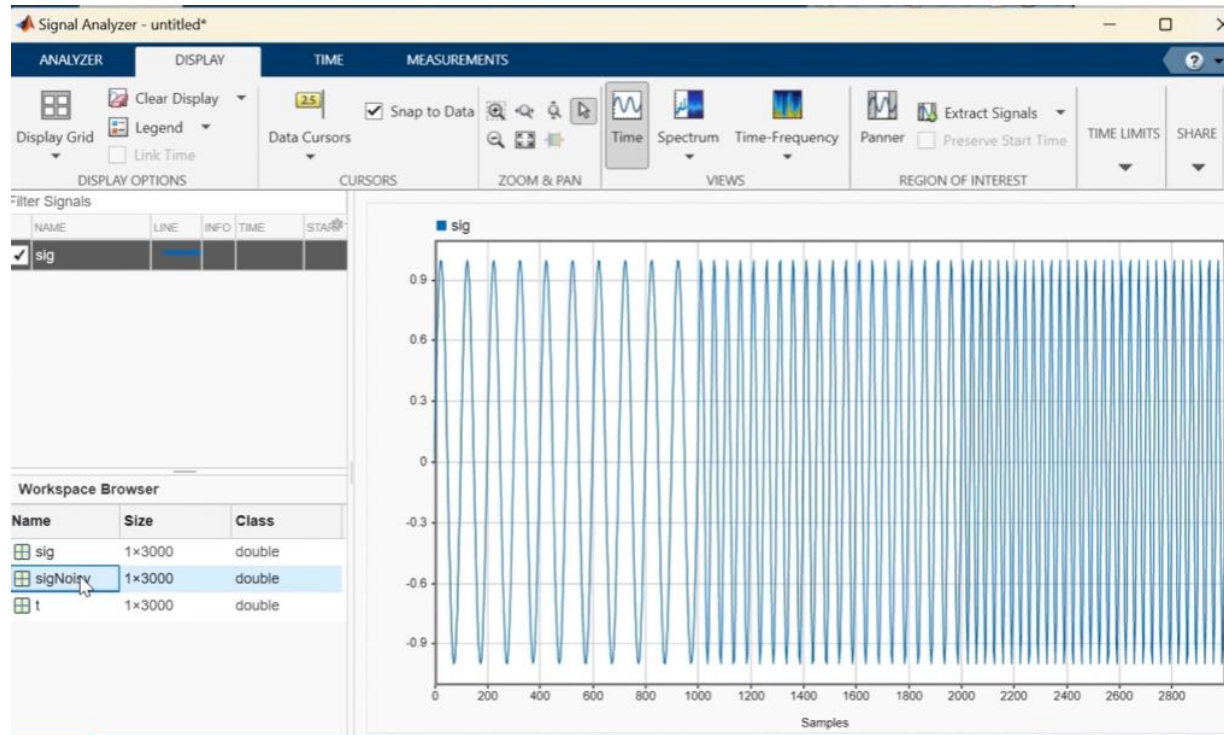
Spectrum Analyzer in MATLAB

is an app for inspecting, comparing and signal processing

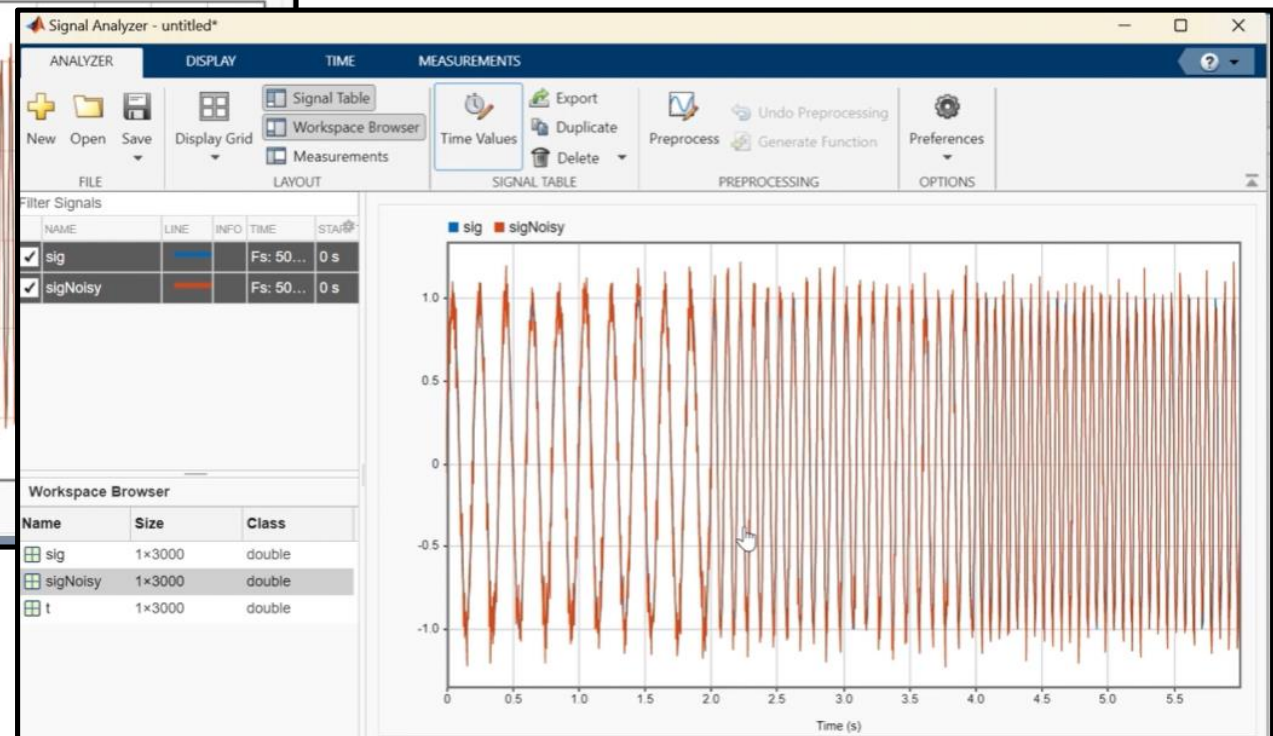
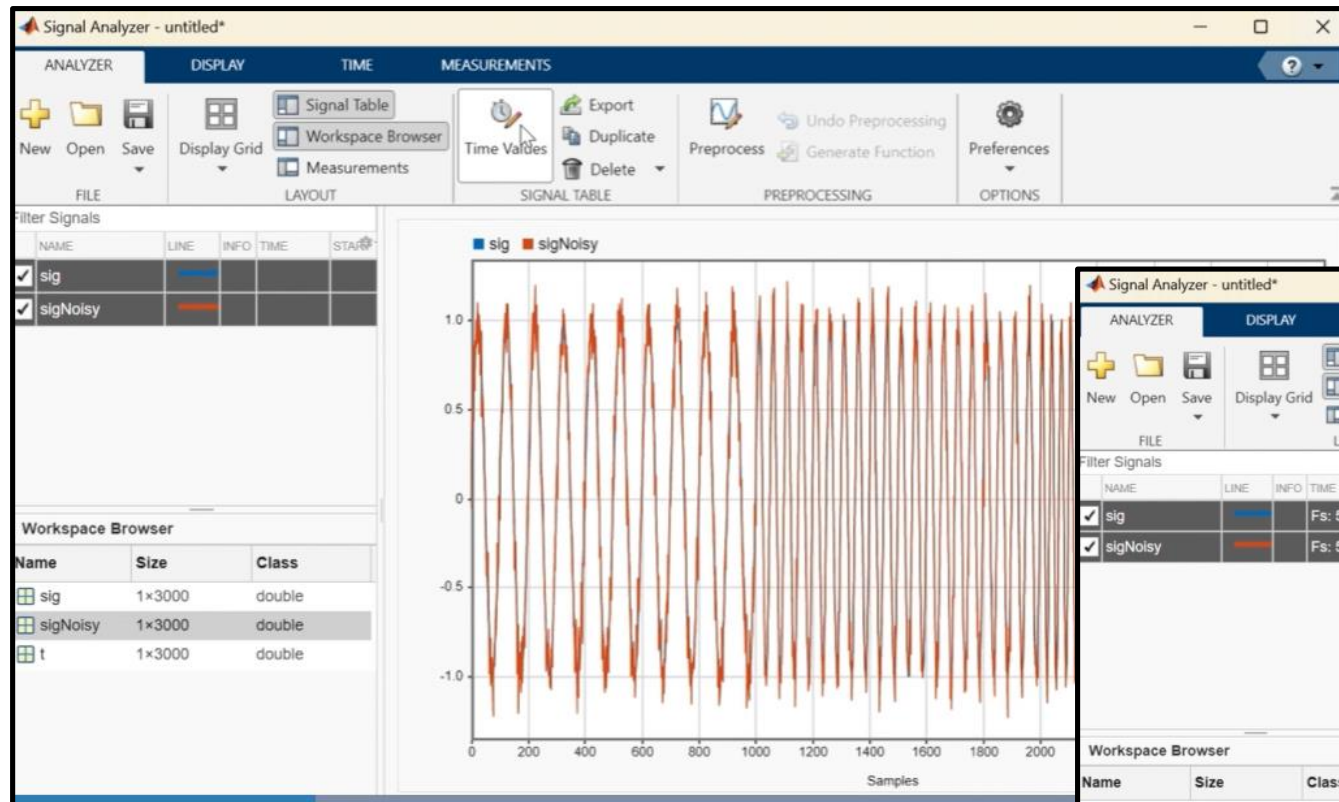
- To open it, click on signal analyzer in the Apps tab
- You can drag signals for the Workspace Browser to the display
- Since these signals are just vectors of numbers, the time information is a number of samples when you add them to the App
- To change the units, select the signals and click Time values as you already have a vector of time values (in the workspace for these signals in the vector t)
- To see the power spectrum of the entire signal, click on spectrum



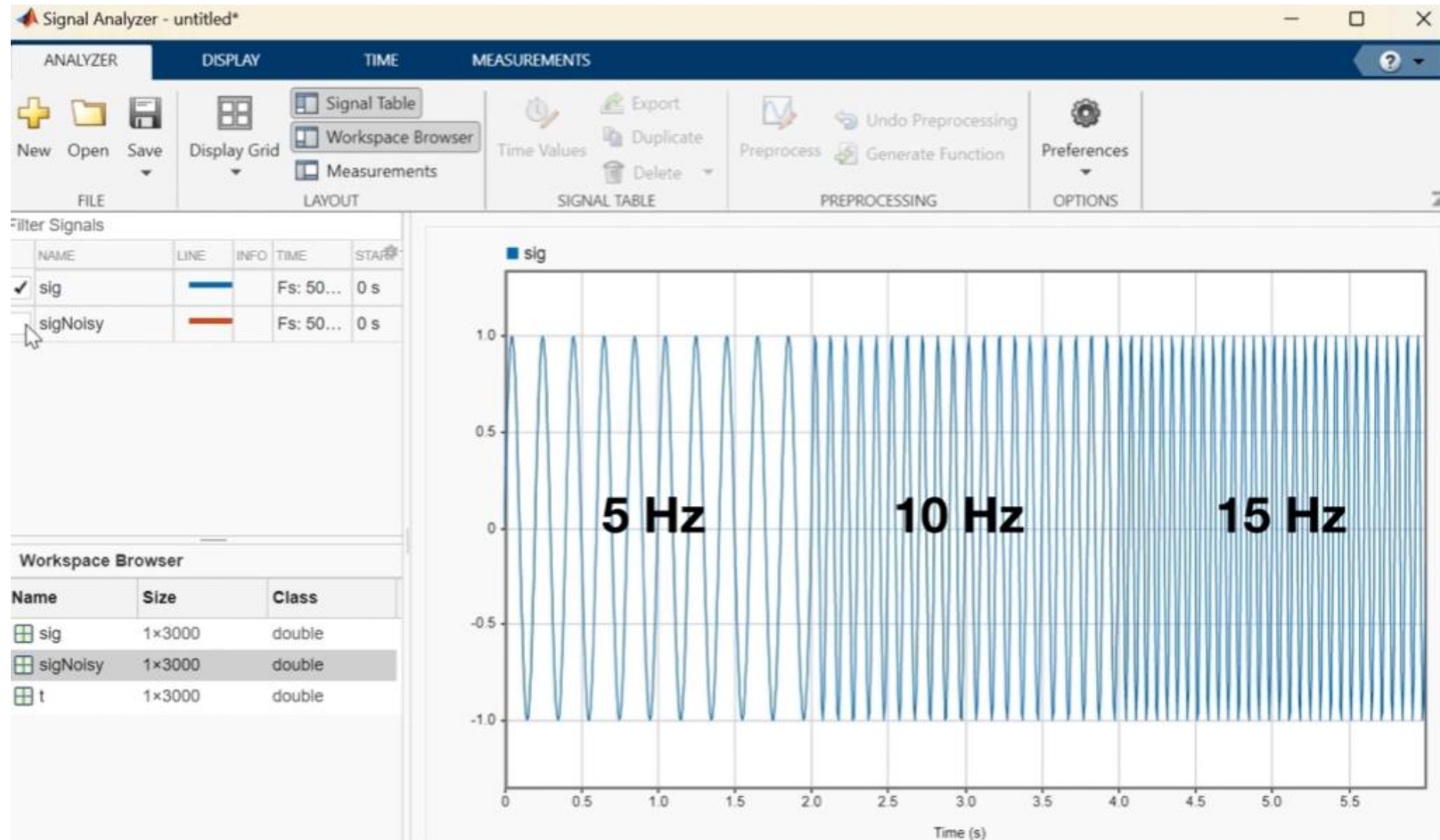
Spectrum Analyzer : Comparing Signals with/without noise



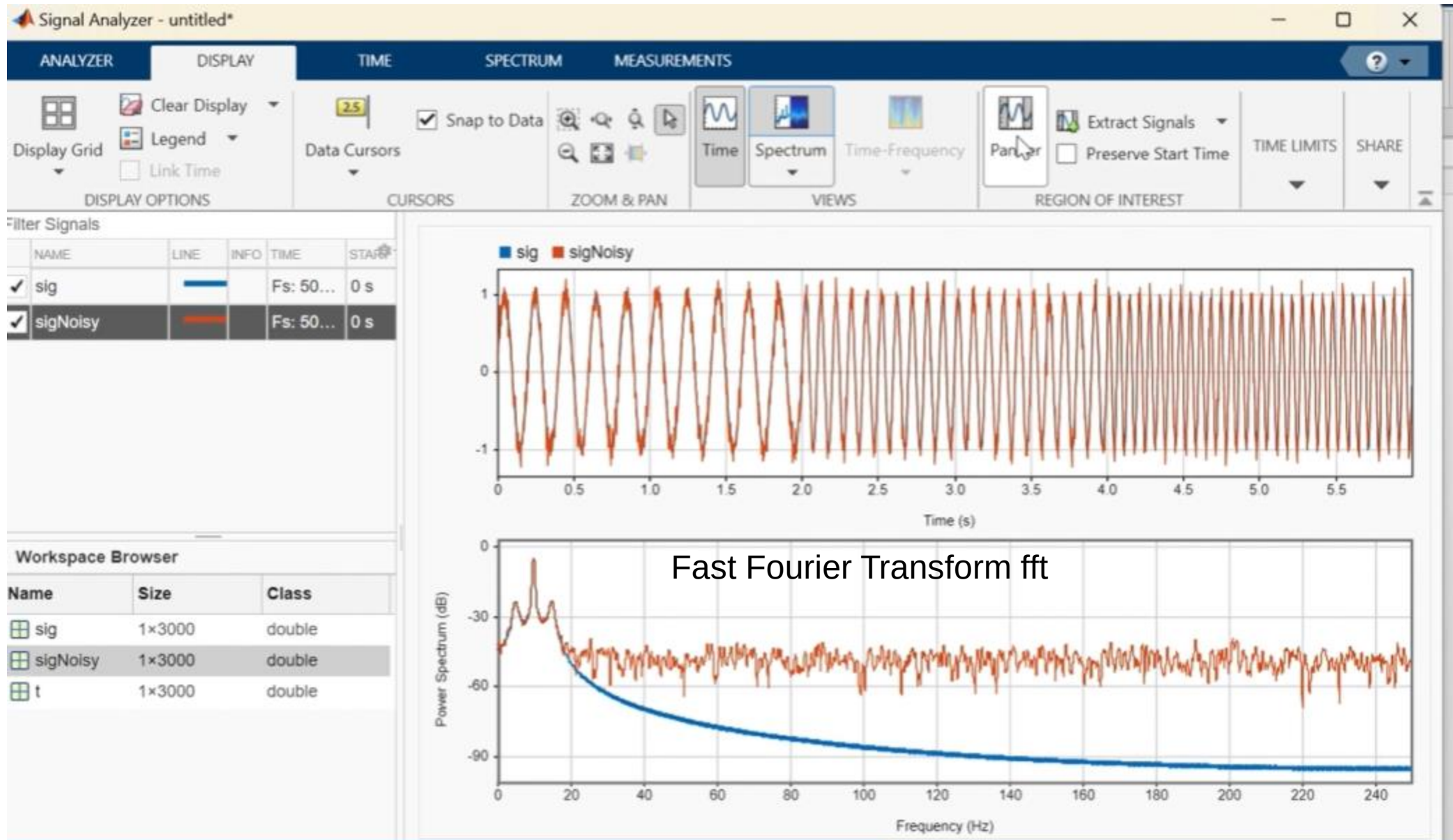
Spectrum Analyzer : Comparing Signals with/without noise



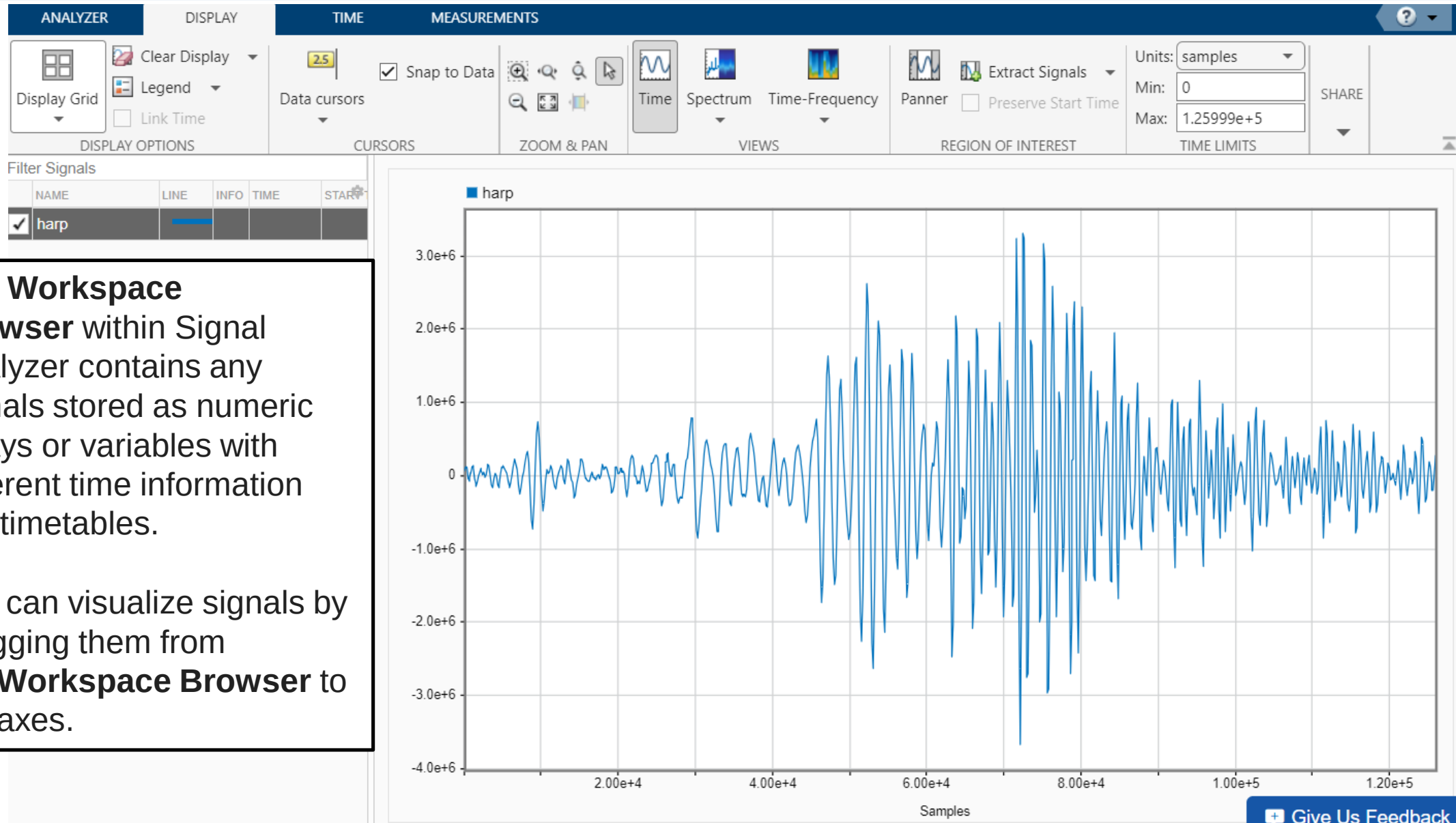
Signals are generated with three different frequencies



View signal in spectrum domain



View Seismic Waves in spectrum analyzer



The **Workspace**

Browser within Signal Analyzer contains any signals stored as numeric arrays or variables with inherent time information like timetables.

You can visualize signals by dragging them from the **Workspace Browser** to the axes.

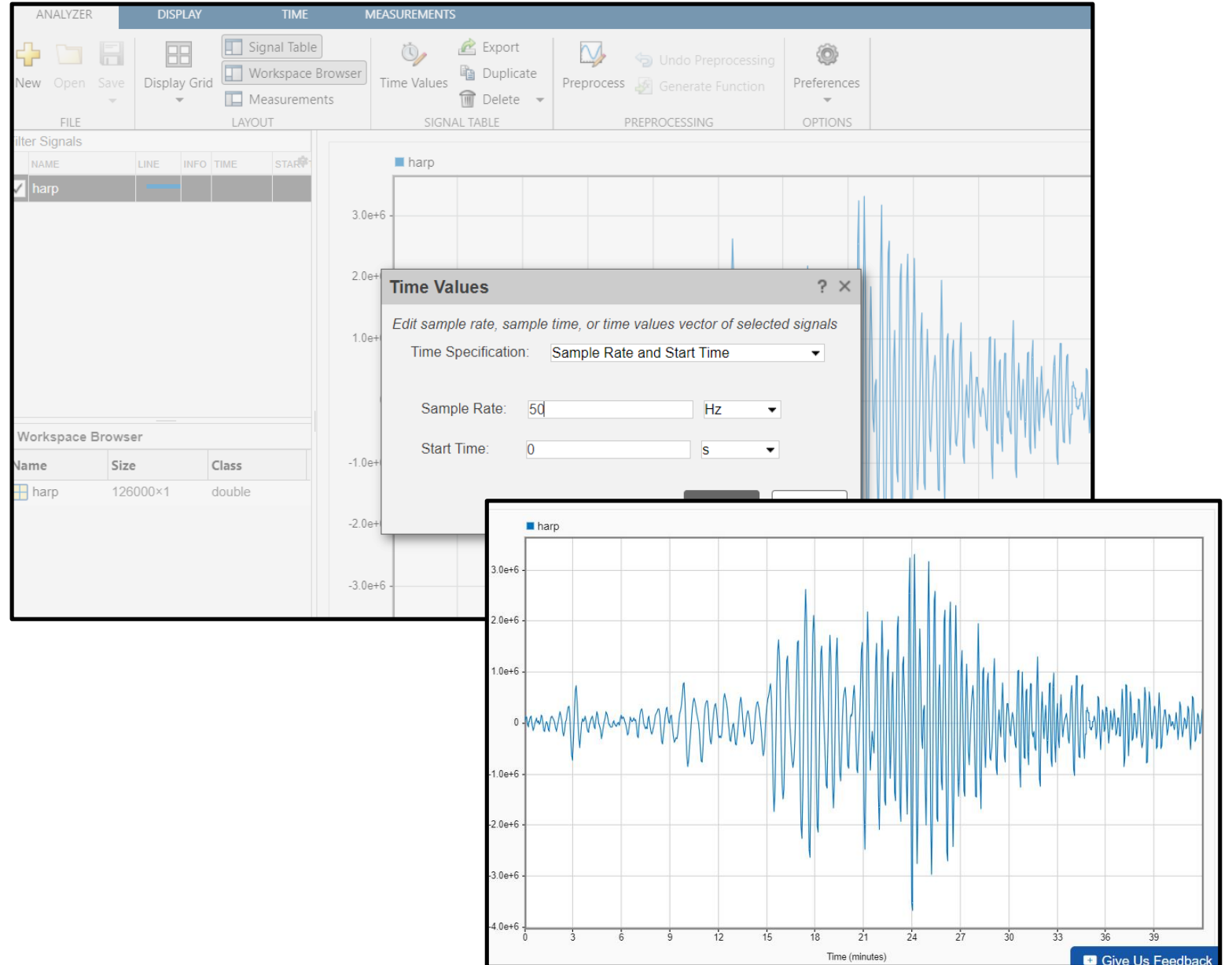
Set time information in spectrum analyzer

To set the time information, select  **Time Values** in the **Analyzer** tab. You can specify the sample rate (f_s), the time between each sample ($\frac{1}{f_s}$), or a vector of time values.

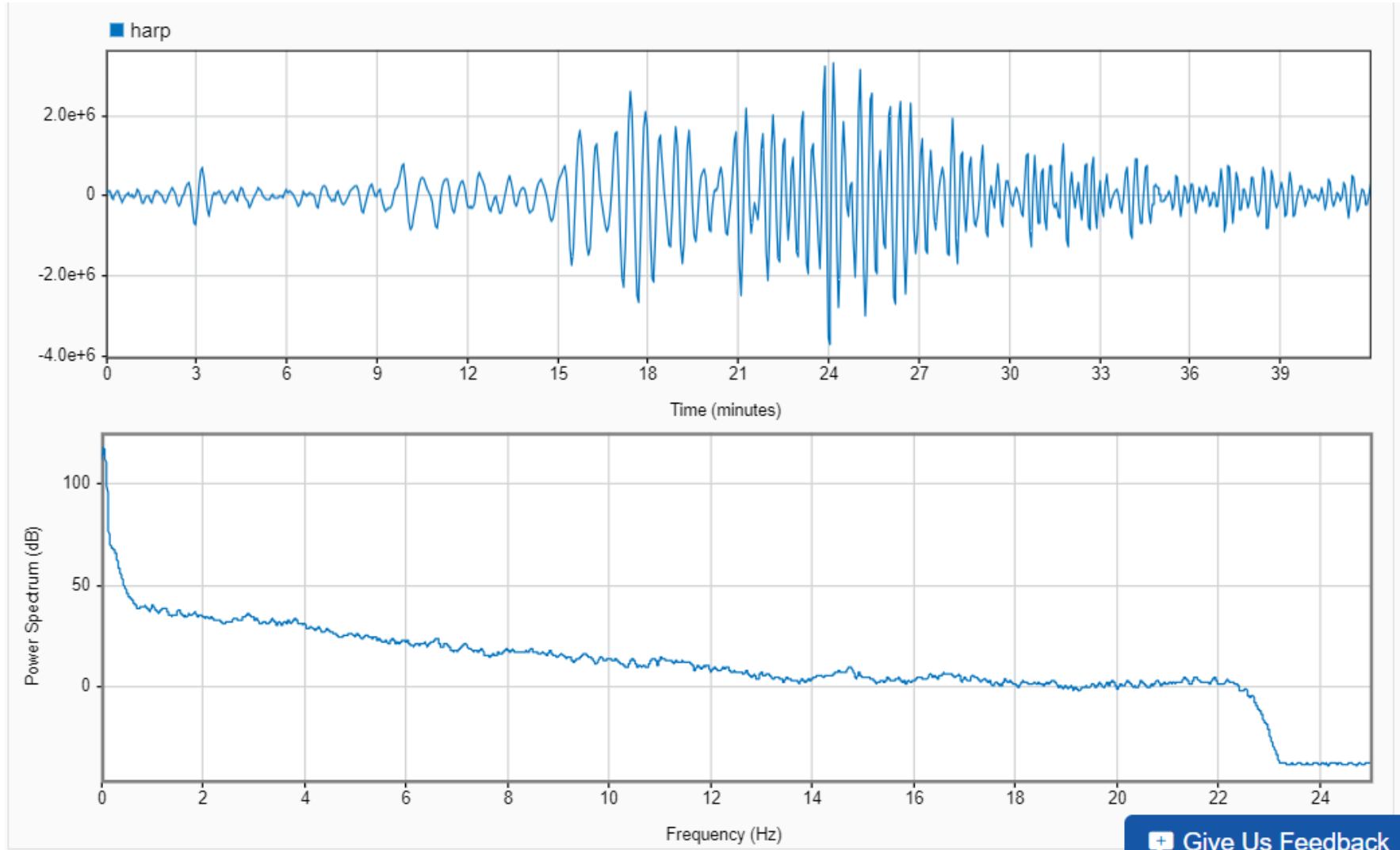
TASK

Set sample rate to Hz.

In the Time Specification drop-down, select Sample Rate and Start Time. Keep the default Start Time of 0 seconds.



To view the frequency content of a signal, you can display its power spectrum by clicking Spectrum on the Display tab



References

- [1] Apolinário, Isabela F., and Paulo SR Diniz. "Introduction to signal processing theory." In *Academic press library in signal processing*, vol. 1, pp. 3-28. Elsevier, 2014.